Arena-Independent Memory for Pure Subgame-Perfect Equilibria in Reachability Games

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In a turn-based game with n players and Boolean reachability objectives, every pure subgame-perfect equilibrium can be replaced by one with exactly the same winner set using at most $n3^n$ memory states per player. The bound holds on every set-sized directed arena without dead ends, with arbitrary branching, and uses vertex updates without storing a previous vertex. Subgame perfection holds after every legal history, including histories inconsistent with the profile, with earlier target visits retained. This answers the arena-independent constrained-memory question posed by Main.

The construction associates finitely many credible payoff labels with each vertex and visited-target set. It retains a compatible payoff promise and otherwise resets it by minimizing the previous owner's payoff bit. A vertex-dependent progress selector fulfills a retained promise without storing a path position or rank. A losing-bit invariant rules out arbitrary unilateral deviations. Combining this compression with the existing lower-semicontinuous-payoff existence theorem yields unconditional pure finite-memory SPE existence on the same arena class. A finite degree-two family with $n = k + 2$ players forces at least $\binom{k}{\lfloor k/2 \rfloor} \geq 2^k/(k+1)$ states in every SPE satisfying a single required winner. The bounds determine exponential-order memory dependence on the player count, without determining the optimal base.

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1 The memory question and the contribution

In a multiplayer graph game, a player chooses the next vertex at each vertex it controls. Its reachability objective asks that a target set be visited at least once. A Nash equilibrium excludes profitable unilateral deviations from the initial history. A subgame-perfect equilibrium (SPE) must do so after every legal history, including histories its prescribed strategies never produce. Thus a succinct equilibrium outcome is insufficient: the strategies also need credible continuations after deviations and after different histories merge at the same vertex.

Main's Automata Exchange question asks whether an SPE satisfying a required subset of reachability objectives, whenever one exists, can be implemented with memory bounded independently of the arena size [Mai25]. We prove a uniform bound, preserving the entire payoff rather than only the required winners. No finiteness or branching assumption is imposed on the arena.

Theorem 1 (Payoff-preserving compression). *Let $n \geq 1$. In every set-sized n-player turn-based directed arena without dead ends, with an initial vertex and Boolean reachability targets, every pure SPE has a pure SPE with exactly the same winner set and at most $n3^n$ vertex-update memory states per player. The players may share their memory set, initial state and update function.*

If an SPE wins for every player in a specified set S, Theorem 1 gives such an SPE with the stated bound, for every S, allowing additional winners. In particular it resolves the full constrained-memory question. The theorem separates internal memory from the size of the graph even when the latter is infinite. Its vertex-dependent strategy tables need not have a finite effective description on an infinite arena.

Corollary 2 (Unconditional existence). *Every game in Theorem 1 admits a pure SPE with at most $n3^n$ vertex-update states per player.*

This corollary combines the compression with an old unrestricted existence theorem of Flesch, Kuipers, Mashiah-Yaakovi, Schoenmakers, Solan and Vrieze [FKMY+10]; it is not a separate existence technique.

Theorem 3 (Supporting lower bound). *For every $k \geq 1$, a finite $(k+2)$-player reachability game with $O(k^2)$ vertices, outdegree at most two, and no cycles except terminal loops admits an SPE winning for a designated player E, but every such SPE requires at least $\binom{k}{\lfloor k/2 \rfloor} \geq 2^k/(k+1)$ vertex-update states for E, irrespective of the other players' memory.*

Consequently the worst-case memory for constrained pure SPEs is $2^{\Theta(n)}$. Exponential SPE memory necessity and its checking mechanisms already have antecedents [BGMR23, FH12]. The lower bound is included to make the quantitative limitation self-contained; no priority claim for exponential necessity or the precise lower-bound constants is made.

1.1 Prior work and the precise advance

The finite-arena results of Brihaye, Bruyère, Goeminne and Raskin [BBGR18, BBGR21] provide the closest reconstruction baseline. In their full version [BBGR18], Proposition 9 identifies weak SPEs with SPEs for reachability, and Proposition 19 and Corollary 21 give arena-dependent payoff-preserving reconstruction through finite symbolic witnesses. Their Theorem 34 already establishes fixed-parameter tractability of constrained existence. We claim no new FPT classification.

A close antecedent is Bruyère, Le Roux, Pauly and Raskin [BLRPR21]. Theorem 7 of their author postprint uses finitely many leaf outcomes and outcome-indexed positional paths to construct finite-memory weak SPEs under reachable absorbing-leaf hypotheses. Theorem 16 gives finite-range weak-SPE existence. General reachability arenas do not satisfy the absorbing-leaf hypotheses: completing all targets in a promised payoff

does not eliminate legal future deviations by the remaining players. We credit outcome memory and positional path selection to this antecedent.

Main [Mai26], Theorem 23 of the author version, gives quadratic player-dependent memory bounds for Nash equilibria in reachability games. Its Remark 1 distinguishes vertex from edge memory. Nash stability alone does not provide credible continuations after every history. Flesch, Kuipers, Mashiah-Yaakovi, Schoenmakers, Solan and Vrieze [FKMY+10], Section 3.3, already use payoff-minimizing continuations after deviations. Alluwaym, Main and Schewe [AMS26] give stationary randomized SPEs on finite graphs and a pure positional special case for absorbing targets; this does not supply prescribed-payoff pure compression on arbitrary arenas.

The advance here is a compatibility and previous-owner normalization rule for all-history SPE payoff labels on continuing arenas. A fixed label is realized by a progress table depending on the current vertex and visited set. A promise is retained whenever it remains available. If it disappears, the previous owner's payoff bit determines a credible reset. Storing that owner requires n states rather than a register ranging over arena vertices. An invariant for every unilateral deviation proves subgame perfection. The inspected primary literature supports this precise distinction; the comparison is bounded and is not an exhaustive priority certificate.

2 Model and memory convention

Let $N = \{1, \dots, n\}$ and let V be a set. An arena consists of a relation $\mathcal{E} \subseteq V \times V$, an owner map $\mathrm{own} : V \to N$, and an initial vertex v_0. Every v has a nonempty set $\mathrm{Succ}(v)$ of successors. Player i has a target $T_i \subseteq V$. Put $t(v) = \{i : v \in T_i\}$. A legal history is a finite nonempty path from v_0; the set of these histories is $\mathcal{H}$. A play is an infinite path. Its payoff set consists of the players whose targets occur at some finite position. We identify a payoff set $p \subseteq N$ with its Boolean vector, writing $p_i = 1$ iff $i \in p$.

A pure strategy chooses a successor after each history ending at a vertex owned by its player. A profile σ fixes a unique continuation after each $h \in \mathcal{H}$. Write $p_\sigma(h)$ for the payoff set of the whole history followed by this continuation, counting their boundary vertex once. In particular target visits before a subgame remain satisfied. A profile is an SPE if, after every $h \in \mathcal{H}$, no player can improve its Boolean payoff by any unilateral strategy change. Deviating strategies can use arbitrary memory and change their moves infinitely often.

The memory convention is

$$M, \quad m_0 \in M, \quad \mathrm{up} : M \times V \to M, \quad \mathrm{next}_i : M \times V_i \to V,$$

where $V_i = \mathrm{own}^{-1}(i)$ and $\mathrm{next}_i(m, v) \in \mathrm{Succ}(v)$. At a history hv, where h may be empty, the strategy plays

$$\mathrm{next}_i(\mathrm{up}^*(m_0, h), v).$$

Thus the incoming memory has processed the prefix strictly before v. The next-move and update functions both receive this incoming state and the current vertex. No previous vertex is available unless explicitly stored. On an infinite arena these functions may be arbitrary tables; only $|M|$ is counted. We work with ordinary set-theoretic choice.

3 Credible labels and uniform progress

Fix the SPE σ to be compressed. For a history h, put $W(h) = \bigcup_{v \text{ in } h} t(v)$ and $x(h) = (\mathrm{last}(h), W(h))$. Let $X = \{x(h) : h \in \mathcal{H}\}$. An augmented state $x = (v, W)$ has, for every $w \in \mathrm{Succ}(v)$, augmented successor $y = (w, W \cup t(w)) \in X$. Define

$$P(x) = \{p_\sigma(h) : h \in \mathcal{H},\ x(h) = x\}.$$

The label set $P(v, W)$ is a nonempty finite subset of 2^N, and each of its labels contains W. It aggregates all legal histories, not just the histories consistent with σ. Neither a witnessing history nor the original strategy's memory will be stored.

Lemma 4 (Local credibility). *For $x = (v, W) \in X$, $p \in P(x)$, $i = \mathrm{own}(v)$, and every augmented successor y of x, some $q \in P(y)$ satisfies $q_i \le p_i$.*

Proof. Choose a history h witnessing $p \in P(x)$ and let w be the vertex component of y. The original continuation after hw gives a label $q = p_\sigma(hw) \in P(y)$. Moving to w at h and thereafter following σ_i is an admissible unilateral change after h. Since σ is subgame perfect, its payoff q_i cannot exceed p_i. This also covers the prescribed successor, for which $q = p$. □

Lemma 5 (Progress without a rank register). *For every $p \subseteq N$, there is a successor table f_p on $X_p = \{x \in X : p \in P(x)\}$ such that, from every $x = (v, W) \in X_p$, following f_p stays in X_p and realizes exactly payoff p, including W.*

Proof. Choose a history witnessing p at x and follow σ. Every augmented state on that continuation is in X_p. Each player in $p \setminus W$ visits its target after finitely many edges. Since there are finitely many players, the maximum of these finitely many visit times is finite (use zero if $p \setminus W$ is empty). Thus some finite path in X_p reaches

$$C_p = \{(u, B) \in X_p : B = p\}.$$

For $x \in X_p$, let $d_p(x)$ be the minimum length of such a path. It is a finite natural number, with no uniform bound asserted. If $d_p(x) > 0$, choose $f_p(x)$ to give a successor $y \in X_p$ with $d_p(y) = d_p(x) - 1$. If $d_p(x) = 0$, choose any successor in X_p, which the witnessing continuation supplies. At rank zero $W = p$, and a successor in X_p has visited set containing W and contained in p, hence equal to p. The selected path therefore reaches rank zero in finitely many steps and thereafter visits no target outside p. Its full payoff is p. The choices are fixed in the vertex-dependent table, not in a memory register. Taking a minimum over natural numbers requires no finite branching. □

4 The vertex-update construction

Let $p_0 = p_\sigma(v_0)$. Use the common memory set

$$M = \{(A, p, j) : A \subseteq p \subseteq N,\ j \in N\}, \qquad m_0 = (\varnothing, p_0, 1).$$

For each player there are three possibilities in the pair (A, p): outside p, inside $p \setminus A$, or inside A. Thus $|M| = n3^n$. The set A records visits strictly before the current vertex, p is an incoming promise, and j is the owner of the previous processed vertex.

Fix a total order on 2^N. For $x \in X$ let $b_j(x)$ be the first label in $P(x)$ minimizing its jth bit. Given $m = (A, p, j)$ and a vertex v, form $B = A \cup t(v)$ and $x = (v, B)$. If $x \in X$, define

$$R(m, v) = \begin{cases} p, & p \in P(x), \\ b_j(x), & p \notin P(x). \end{cases}$$

Writing $r = R(m, v)$, set

$$\mathrm{up}(m, v) = (B, r, \mathrm{own}(v)), \qquad \mathrm{next}_i(m, v) = f_r(x) \quad (v \in V_i). \tag{1}$$

Both functions use the same incoming m: the update does not process a successor, and the next-move function does not receive already updated memory. Since $r \in P(x)$, we have $B \subseteq r$, so the updated state belongs to M. If $x \notin X$, complete both functions arbitrarily with a state in M and a legal successor. These

arguments are unreachable from the initialized machine on a legal history. Call the profile τ. Every player uses the same update, so their memories agree on every history, even when some players have deviated.

Lemma 6 (All-history realization). *At every legal history hv, the incoming A is precisely the visited set of h, and the normalized promise $r = R(m, v)$ belongs to $P(v, A \cup t(v))$. Following τ after hv gives full payoff exactly r.*

Proof. Initially $A = \emptyset$. The update in (1) stores exactly $A \cup t(v)$, independently of the successor taken. Induction over every legal history gives the first assertion and ensures $x \in X$ at each normalization. By definition the normalized label is in $P(x)$.

If the prescribed move $f_r(x)$ is taken, its augmented successor y belongs to X_r by Lemma 5. The incoming promise there is $r \in P(y)$, so normalization retains it. Continuing to follow τ retains r forever and follows f_r. Lemma 5 gives full payoff r, including all visits in the prefix. □

Proof of Theorem 1. Fix any legal history H and player i, and let r be the current normalized promise. By Lemma 6, r_i is the actual payoff under τ after H. If $r_i = 1$, no Boolean improvement is possible. Suppose $r_i = 0$ and allow any unilateral deviating strategy for i. We show that every normalized promise along its continuation still excludes i.

At another player's vertex the prescribed move is taken, retaining the promise at the next vertex. At a vertex owned by i, let y be the augmented state of the successor chosen by the deviation. The update at the current vertex stores previous owner i. If the old promise is in $P(y)$, it is retained. Otherwise the new promise is $b_i(y)$. Lemma 4 supplies some $q \in P(y)$ with $q_i \le r_i = 0$, so the minimizing label also has bit zero. This proves the invariant by induction at every finite position, even with infinitely many deviations.

Every normalized label contains every target visited so far. Hence T_i is never visited on the deviating play, and its payoff is zero. No player can profitably deviate after H. As H was any legal history, τ is an SPE. Finally $p_0 \in P(v_0, t(v_0))$, so the initial normalization retains p_0 and Lemma 6 proves exact initial payoff preservation. The machine has $n3^n$ states, as required. □

5 Unconditional existence from an earlier theorem

Proof of Corollary 2. Theorem 2.3 of [FKMY+10] gives a pure subgame-perfect ε-equilibrium for every $\varepsilon > 0$ in finite-player perfect-information games with bounded lower-semicontinuous payoffs. Its model allows arbitrary action sets (Definition 2.1 and footnote 2), history-dependent feasible actions (Remark 2.1), and pure strategies without measurability constraints (Definition 2.2). Its subgame payoffs retain the prefix (Definition 2.4).

Unfold our arena into its legal-history tree. At a history ending in v, the active player is $\mathrm{own}(v)$ and the available actions are $\mathrm{Succ}(v)$. This is a nonempty set, without a cardinality restriction. One may use V as the common action alphabet and restrict feasible actions by history. The Boolean payoff for visiting T_i is bounded and lower semicontinuous: its one-set is open in the topology generated by finite-history cylinders, because a finite target visit witnesses victory. This holds also for an uncountable discrete alphabet and includes initial target visits. Apply the old theorem with $\varepsilon = 1/2$. A profitable Boolean deviation would improve payoff by one, contradicting its inequality in that subgame. Thus the obtained profile is an exact pure SPE. Theorem 1 compresses it with the claimed memory bound. □

The existence input is essential to this particular deduction and is fully credited to [FKMY+10]. The compression proof itself begins with any given SPE and does not require an algorithm for finding one or computing $P(x)$.

6 A constrained exponential lower bound

We give the complete family for Theorem 3. Its ingredients are related to generalized-reachability subset memory [FH12] and to the credible checking constructions surveyed in [BGMR23, Section 3.1.3]. The argument applies to every SPE satisfying a required winner, not just to profiles carrying a particular prescribed payoff.

Fix $k \geq 1$ and put $r = \lfloor k/2 \rfloor$. The players are $A, E, C_1, \ldots, C_k$. First A selects exactly r indices from $[k]$, giving a set W. Then E selects exactly $k - r$ indices, giving a set U. Each phase examines indices in order and merges all its histories at its end. In particular every r-subset W reaches the same first E vertex.

Here is an explicit implementation. A quota-b phase has decision vertices $q(i, c)$, with $0 \leq i < k$, $0 \leq c \leq i$ and $c \leq b \leq c + k - i$. The two phases use disjoint copies. At $q(i, c)$ the selecting player may choose $d \in \{0, 1\}$ exactly when

$$c + d \leq b \leq c + d + k - i - 1.$$

The edge goes to a forced marker $z(i, c, d)$ and then to $q(i + 1, c + d)$, or to the next phase when $i + 1 = k$. A marker with $d = 1$ is an inclusion of index $i + 1$. Every decision has one or two legal choices. The count is an arena vertex component, not a memory state. There are $O(k^2)$ vertices.

After both phases, play visits checking vertices $c_1, \ldots, c_k$. At c_i, player C_i may go to terminal a or continue to c_{i+1} (to terminal e when $i = k$). Both terminals have only self-loops. The targets of A and E are $\{a\}$ and $\{e\}$, respectively. The target of C_i consists of a and all inclusion markers of index i in either phase. There are no other target memberships. The initial vertex is the first A decision. Ownership at forced vertices is immaterial. The arena is finite, has no dead ends, and has outdegree at most two.

Lemma 7 (Credible terminal outcome). *In any pure SPE, after any legal history reaching the checking phase with $W \cup U \neq [k]$, the continuation ends at a.*

Proof. Otherwise it ends at e, passing every checking vertex. A checker C_i with $i \notin W \cup U$ then receives zero, but can move from c_i to a and receive one. This contradicts subgame perfection at that history. The finite checking chain has no other outcome. □

This lemma does not require the first unsatisfied checker to exit immediately; a later exit could also give it payoff one.

Proof of Theorem 3. An E-winning SPE exists as follows. Player A uses any fixed legal selection. Player E remembers W and, at index i, tries to include i exactly when $i \notin W$; if previous off-profile choices make this choice illegal, it takes any legal choice. At c_i, player C_i continues if its target has already been visited and otherwise goes to a.

After any history in A's phase, all its possible completed choices produce an r-subset, to which E responds with its complement. All checkers then win and continue to e. Player A loses regardless of any unilateral change; all other players win. After a history in E's phase, if its past choices agree with the complement, the unique complementary completion is legal and leads to e. If a past choice disagreed, covering all indices is impossible: an omitted complement index cannot be revisited, while an included index of W consumes one of the exactly $k - r$ response slots. A missing index then causes a checker to exit, so E cannot improve from that subgame. Player A has no further choices. In the checking phase, a satisfied checker already wins, and an unsatisfied checker wins by exiting. A checker whose decision is past cannot affect its continuation. These observations cover arbitrary earlier checking deviations and terminal histories too. Thus the profile is an SPE after every legal history. Its initial outcome is e. It has finite vertex-update memory: E stores W, each checker stores its own visited-target bit, and A is positional.

Now fix any SPE winning initially for E, with no memory restrictions on any player. Its initial outcome is e, so A gets zero. Player A can unilaterally select any r-subset W. Holding all other strategies fixed, the resulting outcome must still give A zero. If the prescribed response U failed to cover $[k]$, Lemma 7, applied to that legal off-profile history, would force a and yield a profitable initial deviation for A. Therefore the single fixed strategy of E must respond to every W by $U = [k] \setminus W$.

Suppose a vertex-update machine represents this strategy of E. Let $m(W)$ be its incoming memory at the common first E vertex after A selects W. If $m(W) = m(W')$ for distinct W, W', determinism forces the same response vertex sequence: the current vertex and incoming state agree, all other moves during this phase are forced, and the update sees the same successive vertices. Hence the response sets coincide. They cannot equal both distinct complements. Thus $W \mapsto m(W)$ is injective, giving at least $\binom{k}{r}$ states. Other players cannot signal through their memories during the response phase. The central binomial coefficient is the largest of $k + 1$ coefficients summing to 2^k, giving the stated inequality. The case $k = 1$ is included; its first quota is zero and its bound is one. □

7 Verification and limits

The proofs establish the arbitrary-arena upper bound and the all-k lower bound; finite computations are supplementary checks. Exact Python best-response checks were applied after every legal-history product state, allowing all edges to generate histories and arbitrary unilateral deviations against fixed opponents. A separate implementation checked 2,827 verified finite SPE inputs and their recompressions, preserving exact initial payoffs; it also checked the lower family for $k = 1, \ldots, 6$. Deliberately omitting the previous-owner register or the progress condition produced failing controls. These tests are neither an exhaustive enumeration of arbitrary games nor formal proof-assistant certification.

Empty targets, initially visited targets, and empty or full payoff sets are included. Finitely many players is used to fulfill a promise after finitely many target visits. No bound on the selector's distance is needed. The result supplies no effective algorithm for the arbitrary infinite tables, no new finite-arena constrained-existence FPT result, and no extension to quantitative arrival-time, Büchi, or arbitrary omega-regular objectives. The gap between the exponential bases of the upper and lower bounds remains.

AI methodology

OpenAI Codex was used substantially in the research and preparation of this article, including proof construction and adversarial checking, the explicit memory machine, the existence-theorem application, the lower-bound family, literature comparison, exact Python verification, and LaTeX drafting. These uses materially affected all technical sections. Additional fresh model contexts checked the mathematical arguments and primary-source comparisons and produced a separate finite checker. Such model checks are not external peer review or formal certification. This disclosure does not assert that the listed author has already performed a separate human verification of every generated argument or reference. Human verification and responsibility are required before any external submission.

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